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Senior Data Engineer

Summary:

- Over **10+** years of experience in Data Engineering, Data Pipeline Design, Development and Implementation as a Sr. Data Engineer/Data Developer and Data Modeler.
- Used **SparkData FramesAPI** over **Cloudera** platform to perform analytics on **Hive** data and Used **Spark** Data Frame Operations to perform required Validations in the data.
- Having extensive knowledge on **RDBMS** such as **Oracle**, DevOps, **Microsoft SQL Server**, **MYSQL**
- Extensive experience working on various databases and database script development using **SQL** and **PL/SQL**.
- Excellent understanding and knowledge of job workflow scheduling and locking tools /services like **Oozie** and **Zookeeper**.
- Good understanding and knowledge of **NoSQL** databases like **MongoDB**, **Azure**, **PostgreSQL**, **HBase** and **Cassandra**.
- Capable of using **AWS** utilities such as **EMR**, **S3** and cloud watch to run and monitor **Hadoop** and **Spark** jobs on **AWS**.
- Developed **Spark** code using **Scala** and **Spark-SQL/Streaming** for faster testing and processing of data.
- Setting up **Azure** infrastructure like **storage accounts**, **integration runtime**, **service principal id**, **app registrations** to enable **scalable** and **optimized** utilization of business user analytical requirements in **Azure**.
- Having good knowledge in writing **Map Reduce** jobs through **Pig**, **Hive**, and **Sqoop**.
- Extensive knowledge in writing **Hadoop** jobs for data analysis as per the business requirements using **Hive** and worked on **Hive QL** queries for required data extraction, join operations, writing custom **UDF's** as required and having good experience in optimizing **Hive** Queries.
- Experienced in building Automation Regressing Scripts for validation of **ETL** process between multiple databases like **Oracle**, **SQL Server**, **Hive**, and **Mongo DB** using **Python**.
- Have good knowledge on **NoSQL** databases like **HBase**, **Cassandra** and **MongoDB**.
- Experienced in working with Amazon Web Services (**AWS**) using **EC2** for computing and **S3** as storage mechanism.
- Hands on experience in using other **Amazon Web Services** like **Auto scaling**, **Redshift**, **Dynamo DB**, **Route53**.
- Experience with operating systems: **Linux**, **Red Hat**, and **UNIX**.
- Expertise in using major components of **Hadoop** ecosystem components like **HDFS**, **YARN**, **Map Reduce**, **Hive**, **Impala**, **Pig**, **Sqoop**, **HBase**, **Spark**, **Spark SQL**, **Kafka**, **Spark Streaming**, **Flume**, **Oozie**, **Zookeeper**, **Hue**.
- Extract Transform and Load data from Sources Systems to cloud **AzureData Storage services** using a combination of **AzureCloud Data Factory**, **T-SQL**, **Spark SQL** and **U-SQL Azure Data Lake Analytics**.
- Experienced on implementation of a log producer in **Scala** that watches for application logs, transform incremental log and sends them to a **Kafka** and **Zookeeper** based log collection platform.
- Experience in writing code in **R** and **Python** to manipulate data for **data loads**, **extracts**, **statistical analysis**, **modeling**, and **data munging**.
- Used **HBase** for storing the **Kafka** topic, partition number and Offsets value. Also used phoenix jar to connect **HBase**table.
- Experience in importing and exporting the data using **Sqoop** from **HDFS** to **Relational Database systems** and **vice-versa** and load into **Hive tables**, which are partitioned.
- Experienced on **Hadoop/Hive** on **AWS**, using both **EMR** and non-**EMR-Hadoop** in **EC2**.

- Knowledge of **ETL** methods for data extraction, transformation and loading in corporate-wide **ETL** Solutions and **Data Warehouse tools** for reporting and data analysis.
- Develop data set processes for **data modelling**, and **Data mining**. Recommend ways to improve data reliability, efficiency and quality.
- Strong experience in writing scripts using **Python API**, **PySpark API** and **Spark API** for analyzing the data.
- Good understanding of distributed systems, **HDFS architecture**, Internal working details of **Map Reduce** and **Spark processing frameworks**.
- Installed and configured **Hive**, **Pig**, **Sqoop**, **Flume** and **Oozie** on the **Hadoop cluster**.

Technical Skills:

- **Big Data Ecosystem:** Map Reduce, HDFS, HIVE 2.3, HBase 1.2 Pig, Sqoop, Impala, Flume 1.8, HDP, Oozie, Zookeeper, Spark, Kafka, storm, Hue Hadoop Distributions Cloudera (CDH3, CDH4, CDH5), Hortonworks
- **Cloud Platform:** Amazon Web Services (AWS), MS Azure
- **Relational Databases:** Oracle 12c, MySQL, MS-SQL Server 2016
- **NoSQL Databases:** HBase, Hive 2.3, and MongoDB
- **Version Control:** GIT, GitLab, SVN
- **Programming Languages:** Java, Python, SQL, PL/SQL, AWS, HiveQL, UNIX Shell Scripting, Scala.
- **Software Development & Testing Life cycle:** UML, Design Patterns (Core Java and J2EE), Software Development Lifecycle (SDLC), Waterfall Model and Agile, STLC
- **Web Technologies:** JavaScript, CSS, HTML and JSP.
- **Operating Systems:** Windows, UNIX/Linux and Mac OS.
- **Build Management Tools:** Maven, Ant.
- **IDE & Command line tools:** Eclipse, Toad.
- **ETL Tools:** DataStage, Informatica.

Work Experience:

Client: CITIZENS BANK Johnston, RI

March 2023 - Present

Role: Senior Data Engineer

Responsibilities:

- Conducted effective requirement analysis, gather requirements and technical rules with business team
- Evaluated analytical data, created complex queries to construct data set for visual analysis and dashboard in **Tableau** using **Teradata** DW tables for business cases and provide insights for data-driven decision making.
- Designed and Developed Real Time Data Ingestion frameworks to fetch data from **Kafka** to **Hadoop**.
- Developed Automation Regressing Scripts for validation of **ETL** process between multiple databases like **AWS Redshift**, **Oracle**, **Mongo DB**, **T-SQL**, and **SQL Server** using **Python**.
- Developed **Java Map Reduce programs** for the analysis of sample log file stored in cluster.
- Loaded data into **HDFS** and extracted the data from **MySQL** into **HDFS** using **Sqoop**.
- Involved in creating **HiveQL** on **HBase** tables and importing efficient work order data into **Hive** tables
- Developed **Map Reduce Programs** for data analysis and data cleaning.
- Exported the patterns analyzed back into **Teradata** using **Sqoop**.
- Imported documents into **HDFS**, **HBase** and creating HAR files.
- Worked in **Agile environment**, and used rally tool to maintain the user stories and tasks.
- Worked on different file formats like Text, Sequence files, Avro, Parquet, ORC, JSON, XML files and Flat files.
- Expertise in **Python** and **Scala**, user-defined functions (UDF) for **Hive** and **Pig** using **Python**.

- Experience in job workflow scheduling and monitoring tools like **Oozie** and good knowledge on **Zookeeper** to coordinate the servers in clusters and to maintain the data consistency.
- Worked with **Enterprise data support** teams to install **Hadoop** updates, patches, version upgrades as required and fixed problems, which raised after the upgrades
- Implemented real time system with **Kafka** and **Zookeeper**.
- Responsible for building **scalable** distributed data solutions using **Hadoop** and involved in Job management using Fair scheduler and Developed job processing scripts using **Oozie workflow**.
- Developed product profiles using **Pig** and commodity UDFs.
- Developed **Spark** Streaming applications to consume data from **Kafka** topics and insert the processed streams to **HBase**.
- Migrated **Map reduce** jobs to **Spark** jobs to achieve better performance.
- Used **Spark-SQL** to Load JSON data and create Schema R DD and loaded it into **Hive** Tables and handled structured data using **SparkSQL**.
- Implemented **Spark** Scripts using **Scala**, **SparkSQL** to access **hive** tables into **Spark** for faster processing of data.
- Good Knowledge and experience in **Amazon Web Service (AWS)** concepts like **EMR** and **EC2** web services successfully loaded files to **HDFS** from **Oracle**, **SQL Server**, **Teradata** and **Netezza** using **Sqoop**.
- Exploratory Data Analysis and Data wrangling wif **R** and **Python**.
- By using **Zookeeper** implementation in the cluster, provided concurrent access for **hive** tables with shared and exclusive locking
- Developed workflow in **Oozie** to automate the tasks of loading the data into **HDFS** and pre-processing with **Pig**.
- Analyzed the **SQL** scripts and designed the solution to implement using **Scala**.
- Created a **Kafka** producer to connect to different external sources and bring the data to a **Kafka** broker.
- Used the **AWS-CLI** to suspend an **AWSLambda** function. Used **AWS CLI** to automate backups of ephemeral data-stores to **S3 buckets**, **EBS**.
- Created a **Kafka** broker in structured streaming to get structured data by schema.
- Working on designing the **Map Reduce** and Yarn flow and writing **Map Reduce** scripts, performance tuning and debugging.
- Developed **Oozie** workflow engine to run multiple **Hive**, **Pig**, **Tealeaf**, **MongoDB**, **Git**, **Sqoop** and **Spark** jobs.
- Installed application on **AWSEC2** instances and configured the storage on **S3** buckets.
- Experience in developing **Map Reduce** Programs using Apache **Hadoop** for analyzing the big data as per the requirement.
- Stored data in **AWSS3** like **HDFS** and performed **EMR** programs on data stored.

Environment: Hadoop, Aws, Kafka, Pig, Hive, Scala, Jdbc, Aws, Sqoop, Zookeeper, Python, Microservice, HDFS, Agile, Scala, Map reduce, Oozie, Python, HBase.

Client: CVS Health. "Scottsdale, AZ

April 2021 - February 2023

Role: Data Engineer

Responsibilities:

- Working as Developer in **hive** and **impala** for more parallel processing data in Cludera systems.
- Working in big data technologies like **spark**, **scala**, **Hive**, **Hadoop** cluster (Cludera platform).
- Implement **SparkKafka** streaming to pick up the data from **Kafka** and send to **Spark** pipeline.
- Good understanding and knowledge of NoSQL databases like **MongoDB**, **Azure**, **PostgreSQL**, **HBase** and **Cassandra**.
- Having good knowledge in writing **MapReduce** jobs through **Pig**, **Hive**, and **Sqoop**.
- Experience in developing customized UDF's in **Python** to extend **Hive** and **Pig** Latin functionality.
- Used Cludera data platform for deploying **Hadoop** in some modules.

- Implemented **OLAP** multi-dimensional cube functionality using **Azure SQL Data Warehouse**.
- Write programs using **Spark** to move data from Storage input location to output location by running data loading, validation, and transformation to the data
- Experience developing **Kafka** producers and **Kafka** Consumers for streaming millions of events per second on streaming data
- Wrote **MapReduce** job using **Pig** Latin. Involved in **ETL**, Data Integration and Migration.
- Working in relational **SQL** and **NoSQL** databases, including Oracle, **Hive**, **Sqoop** and **HBase**.
- Worked with **NoSQL** databases like **HBase** in creating **HBase** tables to load large sets of semi-structured data coming from various sources.
- Schedule nightly batch jobs using **Oozie** to perform schema validation and IVP transformation at larger scale to take the advantage of the power of **Hadoop**.
- Experience in implementing **OLAP** multi-dimensional cube functionality using **Azure SQL Data Warehouse**.
- Closely work on pub-sub model as well because of Lamda model we implemented in tcf bank.
- Design & implement **SparkSQL** tables, **Hive** scripts job with stone branch for scheduling and create work flow and task flow.
- Created and maintained various Shell and **Python** scripts for automating various processes and optimized **Map Reduce** code, **pig** scripts and performance tuning and analysis.
- Experience in Job management using Fair scheduler and Developed job processing scripts using **Oozie** workflow to run multiple **Spark** Jobs in sequence for processing data
- Designed and executed **Oozie** workflows in a manner that allowed for scheduling **Sqoop** and **Hive** job actions to extract, transform and load data
- Data Ingestion to one or more **Azure** Services - (**Azure** Data Lake, **Azure** Storage, **AzureSQL**, **Azure** DW) and processing the data in **Azure** Databricks.
- Designed and implemented **MapReduce**-based large-scale parallel relation-learning system
- Experience in writing **SQOOP** Scripts for importing and exporting data from RDBMS to **HDFS**.
- Experienced in using **Tidal enterprise scheduler** and **Oozie** Operational Services for coordinating the cluster and scheduling workflows.
- Implemented Copy activity, Custom **Azure** Data Factory Pipeline Activities
- Primarily involved in Data Migration using **SQL**, **SQLAzure**, **Azure** Storage, and **Azure** Data Factory, **SSIS**, PowerShell.
- Worked on real time streaming, performed transformations on the data using **Kafka** and **Spark** Streaming
- The custom File System plugin allows **HadoopMap Reduce** programs, **HBase**, **Pig** and **Hive** to work unmodified and access files directly.
- Involved in Importing and exporting data from **HDFS** using **Sqoop**, resolution of access issues, performance issues and Patch/upgrade related issues.
- Perform Big Data analysis using **Scala**, **Spark**, **SparkSQL**, **Hive**, Mlib, Machine Learning algorithms.
- Utilized Apache **Spark** with **Python** to develop and execute Big Data Analytics and Machine learning applications, executed machine Learning use cases under **Spark** ML and Mllib.
- Performing **ETL** testing activities like running the Jobs, Extracting the data using necessary queries from database transform, and upload into the Data warehouse servers.
- Pre-processing using **Hive** and **Pig**.

Environment: HDFS, Azure, Hive, Spark, Linux, Kafka, python, Python, scala, Hadoop, Map Reduce, Zookeeper, Data Bricks, Oozie, Cloudera, Oracle, PL/SQL, Unix, Json

Client: Verizon Irving TX

December 2019 - March 2021

Role: Data Engineer

Responsibilities:

- Working experience on NoSQL databases like **HBase**, **MongoDB** and Cassandra with functionality and implementation.
- Developed data pipelines using **Hive**, **Pig**, and **Map reduce**.
- Aggregated daily sales team updates to send report to executives and to organize jobs running on **Spark** clusters
- Created **Micro services** using **AWS Lambda and API Gateway** using **REST API**.
- Hands on experience in writing **Pig** Latin scripts, working with grunt shells and job scheduling with **Oozie**.
- Implemented Apache Airflow for authoring, scheduling and monitoring Data Pipelines
- Designed several DAGs (Directed Acyclic Graph) for automating **ETL** pipelines
- Performed data extraction, transformation, loading, and integration in data warehouse, operational data stores and master data management
- Played a key role installing and configuring various Big Data ecosystem tools such as Elastic Search, Logstash, Kibana, **Kafka**, and Cassandra.
- Analyzed **Hadoop** cluster using big data analytic tools, including **Kafka**, **Pig**, **Hive**, **Spark**, and **Map reduce**.
- Strong understanding of **AWS** components such as **EC2** and **S3**
- Responsible for data services and data movement infrastructures
- Experienced in **ETL** concepts, building **ETL** solutions and Data modeling
- Worked on architecting the **ETL** transformation layers and writing **spark** jobs to do the processing.
- Worked on various real-time and batch processing applications using **Spark/Scala**, **Kafka** and Cassandra.
- Hands-on experience with large-scale, big data methods including **Hadoop** (worked with components including **HDFS**, **Oozie**), **Spark**, **Hive** (data transformation), **Impala & Hue**
- Compiled data from various sources to perform complex analysis for actionable results
- Measured Efficiency of **Hadoop/Hive** environment ensuring SLA is met
- Developed **Spark** application that uses **Kafka** Consumer and Broker libraries to connect to Apache **Kafka** and consume data from the topics and ingest them into Cassandra.
- Data sources are extracted, transformed and loaded to generate CSV data files with **Python** programming and **SQL** queries.
- Responsible for running **Hadoop** streaming jobs to process terabytes of xml 's data, utilized cluster co-ordination services through **Zookeeper**.
- Worked on data pre-processing and cleaning the data to perform feature engineering and performed data imputation techniques for the missing values in the dataset using **Python**.
- Participated in the full software development lifecycle with requirements, solution design, development, **QA** implementation, and product support using Scrum and other Agile methodologies
- Analyzed the system for new enhancements/functionalities and perform Impact analysis of the application for implementing **ETL** changes
- Implemented a Continuous Delivery pipeline with Docker, and **Git** Hub and **AWS**
- Built performant, **scalable ETL** processes to load, cleanse and validate data.
- Built continuous **Spark** streaming **ETL** pipeline with **Spark**, **Kafka**, **Scala**, **HDFS** and **MongoDB**.
- Installed **Hadoop**, **Map Reduce**, **HDFS**, and **AWS** and developed multiple **Map reduce** jobs in **PIG** and **Hive** for data cleaning and pre-processing.

Environment: AWS, Cloud Sql, MySQL, Postgres, SQLse Server, Kafka, Python, Scala, Spark, Hive, Spark-Sql, Hadoop, Pig.

Client: Robinhood Menlo Park, CA

January 2018 to November 2019

Role: Data Engineer

Responsibilities:

- Build **ETL/ELT** pipeline in data technologies like Py**Spark**, **hive**, presto and **data bricks**
- Headed negotiations to find optimal solutions with project teams and clients

- Mapped client business requirements to internal requirements of trading platform products
- Primarily involved in front-end UI using **HTML5, CSS3, JavaScript, jQuery, and AJAX**.
- Used struts framework to build **MVC** architecture and separate presentation from business logic.
- Utilized various utilities like Struts Tag Libraries, JSP, **JavaScript, HTML, & CSS**.
- Used **Python** & SAS to extract, transform & load source data from transaction systems, generated reports, insights, and key conclusions.
- Built **PL/SQL** (Procedures, Functions, Triggers, and Packages) to summarize the data to populate summary tables that will be used for generating reports with performance improvement.
- Wrote **SQL-Overrides** and used filter conditions in sources qualifier thereby improving the performance of mapping.
- Worked on **OLAP** Data warehouse, Model, Design, and Implementation.
- Monitored performance and optimized **SQL** queries for maximum efficiency.
- Manipulated and summarized data to maximize possible outcomes efficiently
- Involved in creating queues, MDB, Worker to accommodate the messaging to track the workflows
- Optimize data structures for efficiently querying **Hive** and Presto views

Environment: Hadoop, Map Reduce, HDFS, Pig, Hive QL, MySQL, UNIX Shell Scripting, Tableau, Java, Spark, SSIS.

Client: CMC Limited, Hyderabad, India

July 2013 - September 2017

Role: Hadoop Developer

Responsibilities:

- Research and recommend suitable technology stack for **Hadoop** migration considering current enterprise architecture.
- Responsible for building **scalable** distributed data solutions using **Hadoop**.
- Experienced in loading and transforming of large sets of structured, semi-structured and unstructured data.
- Developed **Spark** jobs and **Hive** Jobs to summarize and transform data.
- Involved in converting **Hive/SQL** queries into **Spark** transformations using **Spark** data frames, **Scala** and **Python**.
- Experienced in developing **Spark** scripts for data analysis in both **python** and **Scala**.
- Implemented **Spark** using **Scala** and utilizing Data frames and **SparkSQLAPI** for faster processing of data.
- Handled importing data from different data sources into **HDFS** using **Sqoop** and performing transformations using **Hive** and then loading data into **HDFS**.
- Exporting of a result set from **HIVE** to **MySQL** using **Sqoop** export tool for further processing.
- Collecting and aggregating large amounts of log data and staging data in **HDFS** for further analysis.
- Experience in managing and reviewing **Hadoop** Log files.
- Used **Sqoop** to transfer data between relational databases and **Hadoop**.
- Worked on **HDFS** to store and access huge datasets within **Hadoop**.
- Good hands-on experience with **GitHub**.

Environment: Cloudera Manager, HDFS, Sqoop, Pig, Hive, Oozie, Kafka, Flume, Java, Git.